

Embedded software development companies in India have created a strategic advantage for themselves.

—Ganesh Guruswamy, vice president and country manager, Freescale Semiconductor



ment companies in India have created a strategic advantage for themselves. The success stories of large Indian software services companies like Wipro Technologies, Tata Consultancy Services, HCL Technologies and Mindtree in working with the global design ecosystem underscore our country's strengths in the embedded software space. These companies are now at a stage where they can both conceive and develop products for the global and Indian market," says Ganesh Guruswamy, vice president and country manager, Freescale Semiconductor.

The total embedded software market was worth \$5.9 billion in 2008 and employed 125,600 technical people. Embedded software contributed approximately 81 per cent to the overall revenues and 82 per cent to the overall workforce of the Indian semiconductor industry.

The embedded software sector can be classified into captive and non-captive units. The captive embedded software companies include semiconductor companies that follow the IDM or fabless model and develop the software required for chips and reference boards. OEMs also develop software for their own products. The non-captive companies are third-party embedded software developers, hardware/board designers or the third-party VLSI design companies that provide embedded software solutions as part of their service offerings. There are also a few EMS and ODM companies that provide embedded software development services specific to the products

that they design and manufacture for clients.

The embedded software market is fragmented, with several players operating in India. Embedded software is required by varied industries in the non-traditional areas of discrete manufacturing and industrial process control as well as common industries like automotive electronics, computing products and consumer electronics.

Moving up the value chain

Vivek Sharma, vice president-emerging markets region and director-India Design Center, STMicroelectronics, says that if we talk about our place in the value chain for an embedded system, Indian companies today are more of system solution/product implementers. As most of the users aren't in India, outside teams define what is needed next. In future, though, we can move up the value chain, by migrating from the role of product implementer to product initiator. As the local market grows in coming years, this will create a big opportunity in the local context. Engineers here can become product initiators (deciding what product is to be done), which will give further boost to innovations. And when we innovate, someone has to conceive the ideas. Universities can play a big role here.

The shift has already started: The embedded systems industry in India is undergoing a gradual change from a software-driven outsourcing model to one increasingly focused on growth and development of the domestic market. Indian companies are migrating

from being centres of resource augmentation to product innovation and design centres.

"The last few years have seen a huge growth in the number of small and innovative companies doing complete design of innovative embedded products. A similar trend exists in the captive design centres of multinationals and Indian service companies. They are becoming accountable for more and more of complete product design/delivery/ownership," informs Praveen K. Ganapathy, director-DSP business development, Texas Instruments India.

Access to local customers driven by domestic consumption will enhance product knowledge and further catalyse maturation of the Indian companies to be able to do complete product development, drive product strategies and roadmaps, and do high-end consulting.

The embedded software market will expand as non-captive companies move up the value chain to offer more high-value activities around middleware and develop applications for them.

Market size and forecasts

According to an ISA-Frost & Sullivan report published earlier, the semiconductor and embedded design industry (including chip design, hardware board design and embedded software) in In-

Impact of recession

We cannot ignore the fact that the industry is right now in the midst of a slowdown. Customers are not yet cancelling contracts, but they are taking longer to decide and discretionary spends have slowed. Globally, the total number of design starts is dropping. This has led to a drop in the outsourcing opportunities. In turn, this means there is a cost pressure on the outsourcing.

However, there is a positive side too: Once the market starts bouncing back, companies will first look at outsourcing and then increasing their own headcount. It could mean an opportunity for the Indian design service companies.